

What Falsification Actually Requires

The Structural Consequences of the Standard Science Claims but Does Not Practice

Registry: [\[HOWL-CULT-4-2026\]](#)

Series Path: [\[HOWL-BODY-1-2026\]](#) → [\[HOWL-CULT-1-2026\]](#) → [\[HOWL-CULT-2-2026\]](#) → [\[HOWL-CULT-3-2026\]](#) → [\[HOWL-CULT-4-2026\]](#)

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I. ABSTRACT

This paper states what the institution of science's own standard — falsification as the defining method — actually requires when practiced without softening. The requirements are not new proposals imposed from outside. They are the logical consequences of the standard the institution already claims. The institution teaches that one counterexample kills a universal claim. The institution does not practice this. It patches, extends, adds parameters, and adds dimensions. The standard and the practice have diverged. This paper describes the standard.

Practiced honestly, falsification requires: null results published as first-class findings equal in status to positive results. Falsification chains — sequences in which claims are made, tested, killed, and replaced — celebrated as the mechanism by which science accelerates. Careers built on boundary-finding rewarded at the same level as careers built on territory-claiming. Falsified claims killed without patching — no epicycles, no inserted parameters, no added dimensions to save a framework that failed. Explicit publication of derivation boundaries — every point where a theory stops deriving and starts measuring, published as a research target. Open access to the full record as a structural prerequisite for external falsification. And a decision about scope: either the institution pursues omni-domain investigation and builds the infrastructure for it, or it declares domain-only scope and forfeits authority over the cross-domain territory it has chosen not to occupy.

Each of these requirements follows from the standard the institution teaches. None require external justification. They require only that the institution do what it says it does.

II. THE STANDARD AS TAUGHT

Every student of science learns the same foundational principle. A claim is scientific if and only if it can be shown to be wrong. The method is not confirmation. The method is attempted falsification that the claim survives. One counterexample is sufficient to kill a universal claim. This is the demarcation criterion. It separates science from everything else.

The principle is taught in every introductory course. It appears in every philosophy of science textbook. It is invoked whenever the institution needs to distinguish itself from pseudoscience, from religion, from folk knowledge, from the independent investigator working outside the credentialed system. The institution says: we are different because we submit our claims to falsification. Our claims can be killed. That willingness to be killed is what makes them scientific.

The standard is clear. A universal claim lives until it meets one counterexample. When it meets that counterexample, the claim dies. Not “the claim is modified.” Not “the claim is extended to accommodate the counterexample.” Not “the claim adds a parameter to absorb the discrepancy.” The claim dies. The counterexample is the finding. The death of the claim is the contribution. The next claim, informed by the boundary the counterexample revealed, stands on better ground.

This is the standard. The institution teaches it. The institution invokes it. The institution defines itself by it. The remainder of this paper asks what this standard requires when applied honestly and without exception.

III. THE STANDARD AS PRACTICED

The standard as practiced does not match the standard as taught.

In practice, one counterexample does not kill the claim. The claim is patched. The patch takes various forms depending on the field, but the structure is consistent: a failure is encountered, and instead of killing the claim, the claim is modified to accommodate the failure.

The Standard Model of particle physics has 19 free parameters. Each parameter is a point where the theory does not derive what it needs and a measured value is inserted instead. Each insertion is a patch. The theory predicts with extraordinary precision when 19 measured values are provided. It does not derive those values. Each underived value is a boundary between what the theory explains and what it describes. Those 19 boundaries are not published as boundaries. They are treated as features of the theory — inputs that the theory requires, not failures that the theory has not resolved.

String theory encountered the failure that its predictions did not match observed reality in four dimensions. The response was not to kill the framework. The response was to add dimensions — first 10, then 11, then a landscape of 10^{500} possible solutions. Each added dimension is a patch. The patches made the framework compatible with any observation, which made it unfalsifiable, which by the institution's own taught standard means it is not

longer science. The framework was not killed. It was not retracted. It was quietly deprioritized after decades of funding and careers built on a framework that the institution's own demarcation criterion excludes.

Epicycles are the historical precedent. The Ptolemaic model predicted planetary positions. When predictions failed, circles were added to circles. Each epicycle was a patch that improved the fit without improving the explanation. The model survived for centuries through accumulated patches. It was eventually replaced by a framework — heliocentric mechanics — that did not need the patches. The replacement was resisted for decades because the patched model had institutional investment, credentialed defenders, and a track record of adequate prediction within its accumulated corrections.

The institution teaches this history as a cautionary tale. The institution does not apply the lesson to its own current frameworks. The 19 free parameters are today's epicycles. The extra dimensions are today's epicycles. The measured values inserted where derivations should be are today's epicycles. The institution recognizes epicycles in the past. It does not recognize them in the present.

IV. WHAT PATCHING COSTS

Each patch has a specific cost beyond the complexity it adds. Each patch obscures the boundary that the failure revealed. The failure contained information. The failure said: here is where this framework stops working. That boundary is the most valuable information the framework has produced, because it tells the next investigator exactly where to look.

When the boundary is patched over — when a parameter is inserted, a dimension is added, a correction factor is applied — the boundary disappears from view. The next investigator sees a framework that works (with its patches) rather than a framework that failed at a specific, identifiable point. The patch converts a research target into a solved problem. The problem was not solved. It was covered.

The 19 free parameters of the Standard Model are 19 research targets. Each one is a point where the theory says "I cannot derive this — I need you to measure it and hand it to me." Each one is a signpost pointing at the boundary between explanation and description. If those 19 boundaries were published as boundaries — as explicit admissions of incompleteness at specific, identifiable points — they would constitute the most precise research program in physics. They would say: here are the 19 places where our best theory fails. Solve any one of them and you have advanced fundamental physics.

Instead they are treated as inputs. The theory needs them. The theory works when it has them. The fact that the theory needs them rather than producing them is not treated as a failure. It is treated as a feature of the current state of knowledge. The boundary is absorbed. The research target is hidden. The next investigator sees a working theory rather than a theory with 19 identified failure points.

String theory's extra dimensions are the same cost at larger scale. The framework failed to

predict observed reality in four dimensions. The failure was a boundary: this framework does not work in the dimensionality we observe. That boundary pointed directly at the problem. The added dimensions covered the boundary. The framework now “works” in higher dimensions that cannot be observed, tested, or falsified. The failure was not hidden by malice. It was hidden by the institutional preference for keeping frameworks alive over publishing where they die.

The cost of patching is the cost of the lost map described in [HOWL-CULT-3-2026]. Each patch is a boundary erased from the map. Each erased boundary is a research direction obscured. The cumulative cost is an institution that does not know where its own knowledge fails because the failures have been absorbed into the frameworks through correction factors, free parameters, added dimensions, and inserted measurements. The edges have been smoothed. The map has lost its shape.

V. WHAT A PUBLICATION IS

A publication is a report. It is not a trophy. It is not a claim of territory. It is not a career milestone. It is a report of what happened.

A publication, under the honest standard, contains:

What was done. The method, the observations, the data, the analysis. This is what happened.

What it could mean. The interpretation, the implications, the connections to existing knowledge. This is what the author thinks the results suggest.

How to test it. The specific predictions, the replication conditions, the falsification criteria. This is how the next investigator determines whether the interpretation holds. The publication provides its own kill conditions. It tells the reader exactly what would constitute a counterexample.

What opens up if it is true. The research directions, the applications, the connections to other domains that become available if the claim holds. This is the constructive path forward.

What opens up if it is false. The research directions, the boundary conditions, the alternative explanations that become available if the claim fails. This is the equally constructive path forward in the other direction. The failure of the claim is not a dead end. It is a fork. It tells the next investigator that this direction is closed and other directions, previously blocked by the assumption that this claim was true, are now open.

Both directions are information. Both directions unblock the next investigator. Both directions are contributions. A publication that only describes the positive path — what happens if I am right — is half a report. The other half — what happens if I am wrong, and what does my wrongness reveal — is at least as valuable because it is the half that survives falsification. If the claim dies, the positive path dies with it. The negative path — the map of what the failure reveals — is the permanent contribution.

The current system treats publications as confirmations. The honest standard treats publications as field reports from explored territory. Field reports include the dead ends. Field reports include the places where the investigator turned back. Field reports include the boundaries. A field report that only describes the destinations and omits the terrain is not a map. It is a brochure.

VI. FALSIFICATION AS COLLABORATION

The institution treats falsification as a professional threat. The researcher whose paper is falsified has been “shown to be wrong.” The researcher who performed the falsification has “attacked” a colleague’s work. The language is adversarial. The framing is competitive. The emotional architecture surrounding falsification is hostile.

This is backwards. Falsification is the most productive form of collaboration in science.

Consider the falsification chain. Paper A makes a claim. Paper A says: here is what I found, here is what it means, here is how to test it, here is what opens up if it fails. Paper B tests the claim. Paper B finds a counterexample. Paper B publishes: here is the boundary, here is where Paper A stops working, here is what the failure reveals about the territory. Paper C reads both papers. Paper C now has a map: Paper A explored this territory and found this, Paper B found the boundary of Paper A’s finding, and the boundary reveals a new direction. Paper C explores that direction.

All three papers are contributions. Paper B did not destroy Paper A. Paper B completed Paper A by finding its edge. Paper A mapped territory. Paper B found where the territory ends. Paper C used the boundary to find new territory. The chain is collaborative. Each link depends on the previous links. The falsification is the link that connects the original exploration to the next one.

The researcher who finds the boundary of someone’s claim is that researcher’s most valuable colleague. The boundary is the information the original researcher could not produce because the original researcher was inside the framework looking outward. The falsifier stands outside the framework and sees where it fails. That perspective is the one thing the original researcher cannot have. It is the complementary view that completes the picture.

The institution that treats falsification as professional hostility has turned this collaboration into a threat. The consequence is that researchers do not falsify each other’s work because doing so is socially and professionally costly. The falsification chain — the mechanism by which science accelerates — is broken at the second link because the institution penalizes the second link.

Practiced honestly, falsification chains should be welcomed. Each link unblocks the next. Each falsification reveals a boundary that was invisible from inside the framework. Each boundary opens a direction that was previously closed. The chain is how science moves from where it is to where it needs to be. Breaking the chain at the falsification link stops the movement.

VII. THE CAREER REQUIREMENT

If falsification is the defining method of science, then falsification must be a career.

The current system rewards territory-claiming. The researcher who publishes a novel positive finding is rewarded with citations, tenure progress, grant renewals, and professional recognition. The researcher who publishes a rigorous falsification of an existing finding is not rewarded at the same level. The asymmetry is structural: the institution values new claims over the verification or disconfirmation of existing claims.

The honest standard requires symmetry. Boundary-finding is at least as valuable as territory-claiming. Finding where a theory fails is at least as informative as finding where it works. The researcher who identifies that a landmark paper does not replicate has produced knowledge of equal value to the original paper. In many cases, the falsification is more valuable because it prevents an entire field from building on a foundation that does not hold.

One problem is sufficient. The falsifier does not need to provide a replacement theory. The falsifier does not need to explain why the original claim failed. The falsifier does not need to solve the problem the original claim was attempting to address. The falsification stands alone as a finding. The boundary exists independent of what lies on the other side of it. Requiring the falsifier to provide a replacement before the falsification is accepted is requiring the falsifier to do two jobs — find the boundary and explore beyond it — before the institution will credit the first.

This requirement — that falsification is only valuable when accompanied by a replacement — is the mechanism by which the institution suppresses falsification. It raises the cost of falsification above the cost of original publication. Publishing a new claim requires one finding. Falsifying an existing claim, under the replacement requirement, requires two: the falsification and the alternative. The asymmetry ensures that falsification is always more expensive than confirmation, which ensures that the institutional record accumulates confirmations and sheds falsifications.

The honest standard removes the replacement requirement. You found a problem. One problem. The claim is dead. Publish the problem. The problem is the contribution. What comes next is someone else's paper, or your next paper, informed by the boundary you just found. But the finding of the boundary is complete in itself. It does not need a replacement to be valuable. It needs only to be correct.

VIII. THE COMPLETENESS REQUIREMENT

If a theory cannot derive a value it needs, the theory is incomplete at that point. The point of incompleteness is a boundary. The boundary should be published as a boundary.

The honest standard requires explicit publication of every point where a theory stops deriving and starts measuring. Each such point is a statement: "Here is where our explanation ends and our description begins. We do not know why this value is what it is. We know only that it is what we measured."

This is not a demand for a theory of everything. It is a demand for honesty about the current state. A theory that derives 90% of its predictions and measures the remaining 10% is a theory with 10% identified boundaries. Those boundaries are the most productive research targets available because they are the points where the theory itself says “I do not understand this yet.”

The current practice is to treat measured inputs as temporary. The 19 free parameters of the Standard Model are described as values that will eventually be derived from a deeper theory. This “eventually” has persisted for decades. The promise of future derivation functions as a patch: it converts a boundary into a placeholder, a failure into a promissory note. The note is never called in. The boundary remains hidden behind the promise.

The honest standard cancels the promissory notes. If you cannot derive it today, publish it as a boundary today. Do not promise future derivation. Do not treat the measured value as a temporary input. Treat it as what it is: a point where your theory fails to explain and succeeds only in describing. That point is valuable. That point tells the next investigator exactly where the actual problem is. Hiding it behind “in principle derivable” is hiding the map.

The same standard applies at every scale. If quantum mechanics and general relativity contradict each other at the Planck scale, that contradiction is a boundary. Publish it as a boundary. Not as an unsolved problem — an unsolved problem implies that the existing frameworks will eventually solve it. Publish it as a boundary: here is where our two best theories produce different answers, and neither framework can resolve the contradiction from within. That boundary tells the next investigator that the resolution requires something neither framework currently contains. That is the most precise possible statement of where to look.

IX. THE NO-PATCHING STANDARD

If a claim is falsified, the claim is dead. Not patched. Not modified. Not extended with additional parameters. Not rescued with additional dimensions. Dead.

This is the standard the institution teaches. One counterexample kills a universal claim. The institution does not apply this standard to its own frameworks because applying it would require killing frameworks with decades of institutional investment, careers built on their foundations, and entire departments organized around their assumptions.

The cost of applying the standard honestly is high. The cost of not applying it is higher. Every patch is a boundary erased. Every erased boundary is a research direction obscured. Every obscured direction is time, money, and human effort spent maintaining a framework that should have been replaced by whatever the boundary pointed toward.

The no-patching standard does not mean that falsification is the end. It means that falsification is a transition. The old claim is dead. The new claim, informed by the boundary the falsification revealed, stands on better ground because it knows something the old claim did not: where the old claim fails.

The falsification made the new claim possible. The boundary is the gift. The death of the old claim is not a loss — it is the clearing of the path. The old claim was blocking the view. Its patches were covering the boundaries. Its corrections were smoothing the edges. The new claim, built on the honest boundary, does not need the patches because it was built with the boundary already in view.

This is how the Ptolemaic model was eventually replaced. Not by patching the epicycles. By killing the framework and building a new one that did not need epicycles. The new framework — heliocentric mechanics — was simpler, more predictive, and more explanatory precisely because it started from the boundary the old framework had been covering with patches for centuries.

The honest standard says: do not wait centuries. Kill the claim when it fails. Publish the boundary. Build the next thing. The next thing will be better because it starts from known ground rather than from covered ground.

X. THE ACCESS REQUIREMENT

Falsification requires access. You cannot test a claim you cannot read. You cannot replicate a method you cannot examine. You cannot find a boundary in a framework you cannot see.

The honest standard requires that the full scientific record be publicly accessible. Not because open access is an ideological good — though it may be — but because falsification is structurally impossible without access. An institution that claims falsification as its defining method and restricts access to its record has built a structural contradiction into its own foundation.

Publicly funded research was produced with public money. The public funded the grants, the universities, the laboratories, the salaries. The research output belongs to the funding source. The current system, in which private publishers charge the public to access publicly funded research, is an extraction model that is structurally incompatible with the institution's own stated method.

The honest standard does not require that publishers work for free. It requires that publicly funded research be publicly accessible because falsification — the institution's own defining method — requires it. The mechanism by which access is provided is an implementation detail. The requirement is structural: if the record cannot be examined, it cannot be falsified. If it cannot be falsified, the institution cannot claim to practice falsification. The paywall is not a business decision. It is a methodological contradiction.

The independent investigator, the omni-domain researcher, the practitioner working in the gaps between domains — all of these require access to findings from multiple fields simultaneously. The access barriers are multiplicative: each additional field's literature carries additional subscription costs, additional credential requirements, additional institutional gatekeeping. The researcher who needs to hold six domains simultaneously to see a cross-domain connection faces six layers of access barriers. The institution that claims

authority over all knowledge while structurally preventing cross-domain access has made its own claimed authority unfalsifiable from outside.

XI. THE SCOPE DECISION

The institution must make a decision it has deferred for decades. The decision is about scope.

Option one: the institution claims omni-domain scope. It claims that its method — falsification, empirical testing, systematic investigation — applies to all domains and to the connections between them. If the institution claims this scope, then it must build the infrastructure for cross-domain investigation: departments, credentials, funding, journals, and evaluation frameworks for the gaps between existing domains. This is the requirement identified in [\[HOWL-CULT-2-2026\]](#). If the scope is omni-domain, the gaps must be staffed.

Option two: the institution claims domain-only scope. It claims that its method applies within established domains and does not extend to the territory between them. If the institution claims this scope, then it must acknowledge the limitation explicitly. It must state that cross-domain connections, omni-domain phenomena, and boundary problems are outside its scope. It must stop calling investigators who work in those spaces cranks, because those spaces are outside its declared jurisdiction. It must stop claiming authority over knowledge it has declared itself unable to investigate.

The institution currently claims neither option explicitly. It operates with domain-only infrastructure — departments, credentials, funding, journals organized by domain — while claiming omni-domain authority. It calls itself the custodian of all knowledge while structurally preventing investigation of the connections between its own domains. It dismisses cross-domain investigators as unqualified while offering no qualification for cross-domain investigation.

The honest standard requires choosing. Either build the hallways between the rooms, or stop claiming the building covers all of knowledge. Either fund cross-domain investigation and credential cross-domain investigators, or acknowledge that cross-domain territory is outside institutional scope and belongs to whoever works it.

The current position — domain-only infrastructure with omni-domain claims — is not a position. It is a contradiction sustained by the same structural absence that sustains the falsification forfeiture. The institution has not built the infrastructure for cross-domain work, has not built the infrastructure for falsification, and has not acknowledged the absence of either. The honest standard requires acknowledgment. Choose your scope. Build for it. Stop claiming territory you have chosen not to occupy.

XII. THE EXISTENCE PROOF

The CKS investigation, documented in [\[HOWL-INFO-5-2026\]](#), demonstrates that the honest standard can be practiced.

One investigator explored broadly across multiple domains for 45 days. The investigation produced 390 papers. The investigator tested the mathematics mechanically. The mathematics failed. The investigator killed the entire corpus publicly, immediately, and without qualification. Every paper was marked as falsified. The falsification was timestamped and archived. A post-mortem was published identifying the failure mechanism. The methodology was updated. The surviving content — metamathematical arguments with nonzero M-scores — was extracted and published with honest assessment of scope and limitations.

This is what the standard looks like when practiced. Explore. Test. Find the failure. Kill the claim. Publish the boundary. Extract what survives. Update the method. Continue.

The investigator did not patch. The framework failed and was killed, not modified. The investigator did not file the failure. The failure was published as the primary finding — the mechanism of coherence capture described in [\[HOWL-INFO-5-2026\]](#) is a direct product of the failure, not a separate discovery that happened to occur alongside it. The investigator did not protect the investment. Forty-five days of work across 390 papers was killed in 30 minutes by an arithmetic script. The investment was subordinated to the standard.

The institution should recognize this as its own method practiced at the level the institution describes. It is hypothesis, test, falsification, publication, extraction, continuation. It is one counterexample killing the claim. It is the boundary published as the finding. It is the map updated with the new edges.

The institution may not recognize it because the institution does not practice the method at this standard. The institution patches. The institution files failures. The institution protects investments. The institution rewards confirmations and penalizes falsifications. An investigator who practices the taught standard at the taught level looks unfamiliar to an institution that has drifted from its own standard without acknowledging the drift.

The existence proof stands regardless of recognition. The method works. The standard is practicable. The honest version produces a more complete and more reliable record than the institutional version because it includes the failures, the boundaries, and the post-mortems that the institutional version excludes. The institutional version produces a highlight reel. The honest version produces a map.

XIII. CONCLUSION

The institution of science teaches a standard: falsification is the defining method. Claims are scientific because they can be killed. One counterexample is sufficient. The willingness to be wrong is what separates science from everything else.

The institution does not practice this standard. It patches falsified claims rather than killing them. It adds parameters rather than publishing boundaries. It rewards confirmation rather than falsification. It does not publish null results. It does not fund replication. It does not maintain a record of failures. It restricts access to the record it does maintain. It claims omni-domain authority while building only domain-specific infrastructure.

The honest standard requires: null results as first-class publications. Falsification chains celebrated as the mechanism of acceleration. Careers in boundary-finding rewarded equally with careers in territory-claiming. Falsified claims killed without patching. Derivation boundaries published explicitly as research targets. Open access to the full record. And a scope decision — either build for omni-domain investigation or acknowledge domain-only scope and forfeit the gaps.

None of these requirements are external impositions. Each follows from the standard the institution already claims. The institution teaches falsification. These are the structural requirements of falsification. The institution has not met them. The standard has not changed. The practice drifted. The drift was not announced. The motto on the building was not updated.

By the institution's own standard, the institution is not practicing science as it defines the term. By the honest standard — the standard the institution teaches to its students and invokes to distinguish itself from all other forms of knowledge — the path forward is clear. Publish the boundaries. Kill the dead claims. Celebrate the falsifiers. Open the record. Choose your scope. Build the infrastructure your method requires. Or acknowledge that you have chosen a different method and update the motto accordingly.

The standard the institution claims is a good standard. It produced the modern world when it was practiced. It can produce the next stage of knowledge if it is practiced again. The standard does not need to be replaced. It needs to be applied — honestly, completely, and without the institutional softening that has converted falsification from a practice into a slogan.

Edison's 10,000 failures were science. The file drawer is not. The distance between them is the distance between the standard and the practice. The standard is still there. It is waiting to be practiced. The practice requires only the willingness to do what the motto has always said: be willing to be wrong, and say so when you are.

END HOWL-CULT-4-2026

Registry: [[HOWL-CULT-4-2026](#)]

Status: Complete

Domain: Epistemology / Philosophy of Science / Institutional Reform Analysis

Central Argument: The institution's own stated standard — falsification as the defining method — produces specific structural requirements that the institution has not met

Key Requirements: Null results as first-class publications; falsification chains as collaboration; boundary-finding as rewarded career; no patching of falsified claims; explicit

derivation boundaries; open access; scope decision on omni-domain versus domain-only

Structural Finding: Every requirement follows from the standard the institution already claims; no external framework is imposed; the institution is held to its own stated principles

Conclusion: The standard is good; the practice drifted; the drift was not announced; the standard needs to be practiced, not replaced

[[HOWL-CULT-4-2026](#)] What Falsification Actually Requires. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT-4-2026>

[[HOWL-BODY-1-2026](#)] Neural Territory. Github: <https://github.com/ghowland/papers/tree/main/papers/BODY/HOWL-BODY-1-2026>

[[HOWL-CULT-1-2026](#)] The N=1000 Fallacy. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-1-2026>

[[HOWL-CULT-2-2026](#)] The Domain Boundary. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-2-2026>

[[HOWL-CULT-3-2026](#)] The Falsification Forfeiture. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-3-2026>

[[HOWL-INFO-5-2026](#)] Coherence as Information Phenomenon. Github: <https://github.com/ghowland/papers/tree/main/papers/INFO-5-2026>